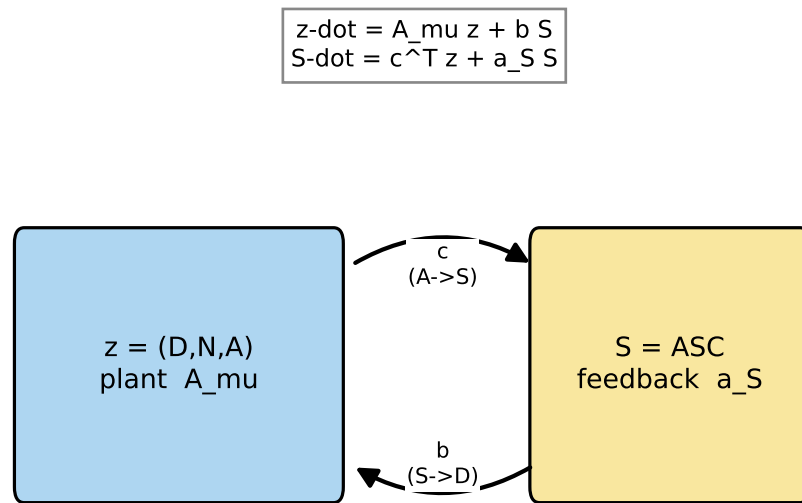


Figure 3 -- Control-theoretic mechanism

A. Mode-resolved feedback block diagram



B. Exact Schur-complement relation

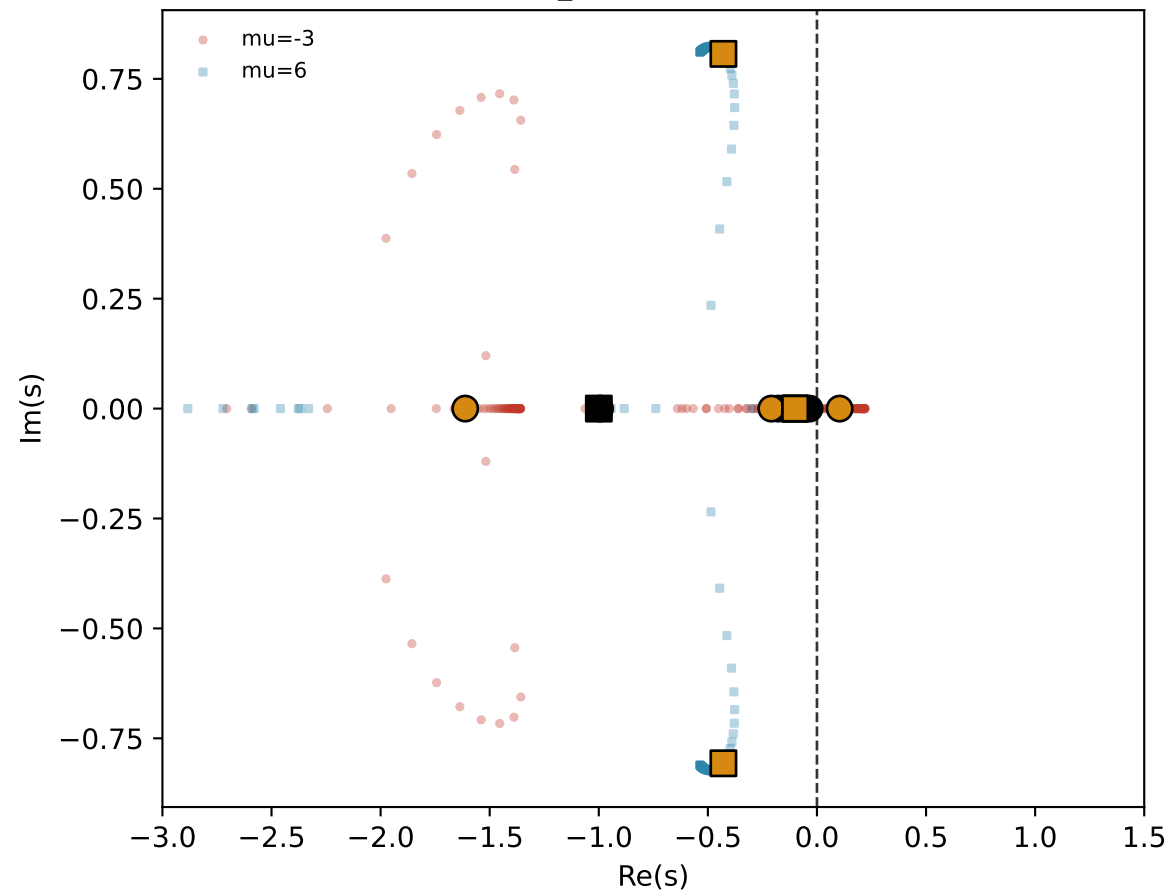
$$P_\mu(s) = c^\top (sI - A_\mu)^{-1} b$$

$$L_\mu(s) = P_\mu(s)/(s - a_S)$$

$$\det(sI - J_\mu) = \det(sI - A_\mu) (s - a_S) [1 - L_\mu(s)]$$

Closed-loop poles $\equiv$ roots of $1 - L_\mu(s) = 0$
 (verified against $\text{eig}(J_\mu)$ to machine precision,
 max coeff. diff. $1.4\text{e-}13$ -- control/schur_verification.py)

C. Root locus vs. lam_N (mu=-3 circles, mu=6 squares)



D. Exact DC loop gain vs. lam_N, vs. G_eff

